

[scale=0.85]  
[font=, black!80] at (11.5,12.8) Why the Optimiser Chose  $n = 12$  Despite Beam-Only Scaling; [font=, shift=(0,0)] [red!3, rounded corners=6pt] (-0.5,-2.0) rectangle (9.0,11.2); [font=, red!60!black] at (4.5,11.2)  $\frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{1 + \frac{1}{n^2} \left( \frac{1}{i^2} + \frac{1}{(i-1)^2} \right)}}$ ; [font=, red!70!black, fill=white, draw=red!30, rounded corners=3pt, inner sep=3pt] at (2.8,8.0)  $n=12$ ; [-i, line width=4pt, i=stealth, magenta!60] (9.4,5.5) -- (11.0,5.5); [font=, magenta!60, align=center] at (10.2,5.5)  $n=12$

system effects;

[shift=(11.5,0)] [blue!2, rounded corners=6pt] (-0.5,-2.0) rectangle (9.5,11.2); [font=, blue!60!black] at (4.5,11.2)  $\frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{1 + \frac{1}{n^2} \left( \frac{1}{i^2} + \frac{1}{(i-1)^2} \right)}}$ ; [-i, thick] (0.8,1.5) -- (0.8,9.0) node[above, font=] Total mass (kg); [-i, thick] (0.8,1.5) -- (9.2,1.5) node[below, font=]  $\frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{1 + \frac{1}{n^2} \left( \frac{1}{i^2} + \frac{1}{(i-1)^2} \right)}}$ ; [font=] at (1.1, 5.5); [blue!60, line width=3pt] plot coordinates (2.2,7.0) (3.8,5.2) (5.8,3.8) (7.4,2.8) (9.0,2.0); [blue!50] (2.2,7.0) (3.8,5.2) (5.8,3.8) (7.4,2.8) (9.0,2.0); [font=, blue!70!black, align=left, fill=white, draw=blue!20, rounded corners=3pt, inner sep=4pt] at (2.2,7.0)  $n=12$

1. Thinner beams at higher  $n$
2. Smaller knuckles (coupled to  $D_o$ )
3. Tether  $\uparrow$  modest, rotor  $\uparrow$  modest

$\rightarrow$  system optimum at  $n=12$ ;

[black!3, rounded corners=6pt] (-0.5,-2.5) rectangle (21.0,-1.5); [font=, black!70, align=center] at (10.2,-2)  $n=12$